§0. Reviews of Analysis 1

· **Sets** (A: a nonempty set)

- **Maximum / Minimum of Sets**
  · $M \in A$ is the maximum of A, if $M \geqslant a \ \ \forall a \in A$.
  · $m \in A$ is the minimum of A, if $m \leqslant a \ \ \forall a \in A$.

- **Bound of Sets**
  · L is an upper bound for A, if $L \geqslant a \ \ \forall a \in A$.
  · l is a lower bound for A, if $l \leqslant a \ \ \forall a \in A$.
  · M is the supremum of A, denoted by $\sup A$,
  if $\exists M = \min\{L : L \text{ is an upper bound for } A\}$.
  · m is the infimum of A, denoted by $\inf A$,
  if $\exists m = \max\{l : l \text{ is a lower bound for } A\}$.

· **Sequences & Series'** ((a_n): a real sequence)

- **The Limit of Sequences**
  : $\lim a_n = a \in \mathbb{R}$
  $\overset{\text{def}}{\iff} \ \forall \epsilon > 0, \ \exists N > 0$ s.t. $n > N$ implies $|a_n - a| < \epsilon$.

- **$\limsup$'s and $\liminf$'s**
  · $\limsup a_n \equiv \lim\limits_{N\to\infty} \sup\{a_n : n > N\}$.
  · $\liminf a_n \equiv \lim\limits_{N\to\infty} \inf\{a_n : n > N\}$.
  · If $\limsup a_n \neq \liminf a_n$, then $\nexists \lim a_n$.

- **Subsequence**
  · (a_{n_k}) is a subsequence of (a_n)
  for an increasing natural sequence (n_k).
  · a is a subsequential limit of (a_n) if $\lim\limits_{k\to\infty} a_{n_k} = a$.
  · $\limsup a_n = \sup A$ and $\liminf a_n = \inf A$
  for $A = \{L : L \text{ is a subsequential limit of } (a_n)\}$.
  $\implies \lim a_n$ is defined iff $|A| = 1$.

- **Cauchy Sequences**
  · (a_n) is a Cauchy sequence if $\forall \epsilon > 0$;
  $\exists N > 0$ s.t. $n > m > N$ implies $|a_n - a_m| < \epsilon$.
  · (a_n) converges iff (a_n) is a Cauchy sequence.

- **Convergence of Series'**
  · $\sum a_n$ converges to S if a sequence (S_n) s.t.
  $$S_n = \sum_{i=m}^{n} a_i \quad (m = \min n) \quad \forall n,$$
  which are "the partial sums of (a_n)," converges to S.
  · If $\sum a_n \in \mathbb{R}$, then $\lim a_n = 0$.

- **Absolute Convergence of Seq.'s & Series'**
  · (a_n) absolutely converges if $(|a_n|)$ converges.
  · If $\sum |a_n| \in \mathbb{R}$, then $\sum a_n \in \mathbb{R}$.

- **Limit Theorems of Seq.'s & Series'**
  · $\lim\limits_{n\to\infty} \dfrac{1}{n^p} = 0$ if $p > 0$.
  · $\lim\limits_{n\to\infty} r^n = 0$ if $|r| < 1$.
  · $\lim\limits_{n\to\infty} \left(1 + \dfrac{r}{n}\right)^n = e^r$.
  · Squeeze Theorem (a_n, b_n, c_n: seq.'s)
  : If $b_n \leqslant a_n \leqslant c_n \ \forall n > N$ and $\lim b_n = \lim c_n = L$,
  then $\lim a_n = L$.
  · $\sum \dfrac{1}{n^p}$ if $p > 1$.

- **Alternating Series Theorem**
  : $\sum |a_n|$ converges if
  (i) $\exists N_1 > 0$ s.t. $i > N_1$ implies $|a_i| > |a_{i+1}|$,
  (ii) $\exists N_2 > 0$ s.t. $j > N_2$ implies $a_j a_{j+1} < 0$,
  (iii) $\lim a_n = 0$.

· **Convergence Tests/Theorems of Series'**

- **Comparison Test** ($a_n \geqslant 0 \ \ \forall n$)
  · $\sum a_n \in \mathbb{R} \wedge |b_n| \leqslant a_n \ \ \forall n \implies \sum b_n$ converges.
  · $\sum a_n = +\infty \wedge b_n \geqslant a_n \ \ \forall n \implies \sum b_n + \infty$.

- **Ratio Test**
  · $\sum a_n$ (abs.) converges iff $\limsup \left|\dfrac{a_{n+1}}{a_n}\right| < 1$.
  · $\sum a_n$ diverges iff $\liminf \left|\dfrac{a_{n+1}}{a_n}\right| > 1$.

- **Root Test**
  · $\sum a_n$ (abs.) converges iff $\limsup |a_n|^{1/n} < 1$.
  · $\sum a_n$ diverges iff $\liminf |a_n|^{1/n} > 1$.

- **Cauchy Criterion**
  : $\sum a_n$ converges iff it satisfies the Cauchy criterion:
  $$\text{Given } \epsilon > 0, \ \exists N > 0 \quad \text{s.t.}$$
  $$n > m > N \quad \text{implies} \quad \left|\sum_{k=m}^{n} a_k\right| < \epsilon.$$

- $\liminf \left|\dfrac{a_{n+1}}{a_n}\right| \leqslant \liminf |a_n|^{\frac{1}{n}}$
  $\leqslant \limsup |a_n|^{\frac{1}{n}} \leqslant \limsup \left|\dfrac{a_{n+1}}{a_n}\right|$.

§23. Power Series

· Power Series

- **Definition**

: a series

$$\sum_{n=0}^{\infty} a_n(x-a)^n$$

with a real sequence $(a_n)_{n=0}^{\infty}$, a variable x, and $a \in \mathbb{R}$. It becomes a function of x if it converges.

· Convergence of Power Series

pf.

For a power series $\sum a_n x^n$, let $b_n = a_n x^n$ for all n, and let $\beta = \limsup |a_n|^{1/n}$.
Then $\limsup |b_n|^{1/n} = |x| \cdot \limsup |a_n|^{1/n} = \beta|x|$, thus for

$$R = \begin{cases} 0 & \text{if } \beta = +\infty, \\ 1/\beta & \text{if } 0 < \beta < +\infty, \\ +\infty & \text{if } \beta = 0, \end{cases}$$

by the Root Test, we have

$$\sum a_n x^n \begin{cases} \text{converges} & \text{if } |x| < R, \\ \text{diverges} & \text{if } |x| > R, \\ \text{has to be checked} & \text{if } |x| = R. \end{cases}$$

■

- We call R the radius of convergence for $\sum a_n x^n$.
- We call $x = \pm R$ the endpoints.
- We call the set of x where $\sum a_n x^n$ converge the interval of convergence for $\sum a_n x^n$.
- **Examples**

· $\sum x^n/n!$ converges for all $x \in \mathbb{R}$.

sol.

Let $a_n = 1/n!$. Then $\beta = \lim \left| \frac{a_{n+1}}{a_n} \right| = 0$, thus $R = +\infty$, and the interval of convergence is $(-\infty, +\infty)$. ■

· $\sum x^n/n$ converges on $[-1, 1)$.

sol.

Let $a_n = 1/n$. Then $\beta = \lim \left| \frac{a_{n+1}}{a_n} \right| = 1$, thus $R = 1$.
Judge endpoints: $\sum (-1)^n/n$ converges, but $\sum 1/n$ diverges.
Therefore, the interval of convergence is $[-1, 1)$. ■

§24. Uniform Convergence

· Pointwise Convergence

- **Definition**

: For a seq. of real-valued ftn.s defined on $S \subseteq \mathbb{R}$: (f_n), if $\exists$ a real-valued ftn. f defined on S s.t.

$$\lim_{n\to\infty} f_n(x) = f(x) \quad \exists \forall x \in S,$$

(f_n) converges pointwise.

· Uniform Convergence

- **Definition**

: For a seq. of real-valued ftn.s (f_n) defined on $S \subseteq \mathbb{R}$, and for a real-valued ftn. f defined on S, if

$$\text{for each } \epsilon > 0, \exists N > 0 \text{ s.t.}$$
$$|f_n(x) - f(x)| < \epsilon \quad \exists \forall n > N \text{ and } \exists \forall x \in S,$$

(f_n) converges uniformly.

- **Continuity of the limit fcn.**

: If each f_n is cont. on $x_0 \in S$ & $f_n \longrightarrow f$ unif. on S, then f is also cont. on x_0.

pf.

Let $\epsilon > 0$. Since $f_n \longrightarrow f$ uniformly on S,

$$\exists N > 0 \text{ s.t. } |f_n(x) - f(x)| < \epsilon/3$$

for all $n > N$ and for all $x \in S$.

And also since every f_n is continuous at $x_0 \in S$,

$\exists \delta > 0$ s.t. $|x - x_0| < \delta$ implies $|f_n(x) - f_n(x_0)| < \epsilon/3$.

Let $n > N$, and $|x - x_0| < \delta$ for $x \in S$. Then, by the triangle inequality,

$$\begin{aligned} &|f(x) - f(x_0)| \\ &= |f(x) - f_n(x) + f_n(x) - f_n(x_0) + f_n(x_0) - f(x_0)| \\ &\leqslant |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)| \\ &< 3 \cdot \epsilon/3 = \epsilon, \end{aligned}$$

which shows that f is continuous at x_0. ■

- **Remark**

: (f_n) on $S \subseteq \mathbb{R}$ converges unif. to f on S iff

$$\lim_{n\to\infty} \sup\{|f(x) - f_n(x)| : x \in S\} = 0.$$

pf.

($\Longrightarrow$) Let $\epsilon > 0$. Then

$\exists N > 0$ s.t. $|f_n(x) - f(x)| < \epsilon \quad \forall n > N, \ \forall x \in S.$

Since, we have

$$\sup\{|f_n(x) - f(x)| : x \in S\} < \epsilon \quad \forall n > N,$$

which implies

$$\lim_{n\to\infty} \sup\{|f_n(x) - f(x)| : x \in S\} = 0.$$

■

§25. More on Uniform Convergence

· Limits of Integrals

: For (f_n) where f_n is real-valued cont. fcn. on $[a, b] \ \forall n$, if $f_n \longrightarrow f$ uniformly, then

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b \lim_{n\to\infty} f_n(x)\,dx = \int_a^b f(x)\,dx.$$

pf.

Since f_n is continuous; integrable on $[a, b]$, and $f_n \longrightarrow f$ uniformly, f is continuous; integrable on $[a, b]$, and thus $f_n - f$ is so.

Since $f_n \longrightarrow f$ uniformly on $[a, b]$, for any $\epsilon > 0$,

$$\exists N > 0 \quad \text{s.t. } |f_n(x) - f(x)| < \frac{\epsilon}{b-a} \quad \forall x \in [a, b], \ \forall n > N.$$

Then we have $n > N$ implies

$$\begin{aligned}\left|\int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx\right| &= \left|\int_a^b (f_n - f)(x)\,dx\right| \\ &\leqslant \int_a^b |(f_n - f)(x)|\,dx \\ &\leqslant \int_a^b \frac{\epsilon}{b-a}\,dx \\ &= \epsilon,\end{aligned}$$

thus we conclude $\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx.$ ■

· Uniformly Cauchy

- **Definition**

: For (f_n) where f_n is a ftn. defined on $S \subseteq \mathbb{R} \ \forall n$, (f_n) is uniformly Cauchy on S, if

$$\text{for each } \epsilon > 0, \ \exists N > 0 \text{ such that}$$
$$|f_n(x) - f_m(x)| < \epsilon \quad \forall x \in S, \ \forall n, m > N.$$

- **Relation with Convergence**

: For (f_n) where f_n is a ftn. defined on $S \subseteq \mathbb{R} \ \forall n$,

(f_n) is uniformly Cauchy on S if and only if $\exists$ a ftn. f s.t. $f_n \longrightarrow f$ uniformly.

pf.

($\Longrightarrow$) Since (f_n) is uniformly Cauchy on S, for each $\epsilon > 0$, and for each $x_0 \in S$, $\exists N_0 > 0$ s.t. $n, m > N_0$ implies $|f_n(x_0) - f_m(x_0)| < \epsilon$.

Then $(f_n(x_0))$ is a Cauchy seq. for each $x_0 \in S$, thus $(f_n(x_0))$ is a convergent seq., which makes us be able to define a fcn. f on S s.t. $f(x) = \lim_{n\to\infty} f_n(x) \ \forall x \in S$.

Since (f_n) is uniformly Cauchy on S, $\exists N > 0$ s.t.,

$$f_n(x) \in (f_m(x) - \epsilon/2, \ f_m(x) + \epsilon/2)$$

$\forall x \in S$, for $\epsilon > 0$ and $\forall n > N$, and fixed $m > N$.

Then, since $f_n \longrightarrow f$ pointwise on S, we can see

$$f(x) \in [f_m(x) - \epsilon/2, \ f_m(x) + \epsilon/2]$$
$$\Longrightarrow |f(x) - f_m(x)| \leqslant \epsilon/2.$$

Thus, we have

$$\begin{aligned}|(f_n - f)(x)| &= |(f_n - f_m)(x) + (f_m - f)(x)| \\ &\leqslant |(f_n - f_m)(x)| + |(f_m - f)(x)| \\ &< \epsilon/2 + \epsilon/2 = \epsilon \quad \forall x \in S,\end{aligned}$$

which implies $f_n \longrightarrow f$ uniformly on S. □

. . . . . .

($\Longleftarrow$) Since $f_n \longrightarrow f$ uniformly on S, for $\epsilon > 0$, $\exists N > 0$ s.t.

$$n > N \text{ implies } |f_n(x) - f(x)| < \epsilon/2 \quad \forall x \in S.$$

Then, for $n > m > N$,

$$\begin{aligned}|(f_n - f_m)(x)| &= |(f_n - f)(x) - (f - f_m)(x)| \\ &\leqslant |(f_n - f)(x) - |(f_m - f)(x)| \\ &< \epsilon/2 + \epsilon/2 = \epsilon \quad \forall x \in S,\end{aligned}$$

which implies (f_n) is uniformly Cauchy on S. ■

· Series of Functions

- Definition of Convergence/Divergence

: For a sequence $(g_k)_{k=0}^{\infty}$ of function on $S \subseteq \mathbb{R}$,

$\sum_{k=0}^{\infty} g_k(x) = \sum g_k$ converges if a sequence (f_n) of partial sums of g_k, i.e., $f_n(x) = \sum_{k=0}^{n} g_k(x)$, converges.

- Continuity of Limit Function

: For a series $\sum g_k$ of functions on $S \subseteq \mathbb{R}$,

if $\left[\begin{array}{l} \text{(i) } g_k \text{ is continuous on } S \text{ for each } k, \\ \text{(ii) } \sum g_k \text{ uniformly converges on } S, \end{array}\right.$

then $\sum g_k$ represents a continuous function on S.

pf.

Let $f_n = \sum_{k=0}^{n} g_k$.

Since g_k is cont. on S $\exists \forall k$, f_n is also cont. on S $\exists \forall n$, and by the assumption, $f_n \longrightarrow f$ uniformly on S for a function f, which is cont., by Thm 24.3.

Thus we have $\sum g_k$ represents a cont. ftn. on S. ■

- Cauchy Criterion

: For a series $\sum g_k$ of functions on $S \subseteq \mathbb{R}$,

$\sum g_k$ converges uniformly on S,

iff $\sum g_k$ satisfies the Cauchy criterion uniformly on S.

pf.

Let $f_n = \sum_{k=0}^{n} g_k$, and apply Thm 25.4 & Exc 25.4. ■

- Weierstrass M-test

: For a seq. (M_k) of nonnegative real number,

and a seq. (g_k) of functions on $S \in \mathbb{R}$,

if $\sum M_k < +\infty$, and $|g_k(x)| \leqslant M_k$ $\exists \forall k$ and $\exists \forall x \in S$,

then $\sum g_k$ converges uniformly on S.

pf.

Since $\sum M_k < \infty$, it satisfies Cauchy criterion on S; given $\epsilon > 0$, $\exists N > 0$ such that

$$n \geqslant m > N \text{ implies } \left|\sum_{k=m}^{n} M_k\right| < \epsilon.$$

Let $n \geqslant m > N$ and $x \in S$. Then

$$\left|\sum_{k=m}^{n} g_k(x)\right| \leqslant \sum_{k=m}^{n} |g_k(x)| \leqslant \sum_{k=m}^{n} M_k < \epsilon,$$

so we have that $\sum g_k$ satisfies Cauchy criterion uniformly on S, thus $\sum g_k$ converges uniformly on S. ■

- If $\sum g_n$ converges uniformly on S,

then $\lim_{n\to\infty} \sup\{|g_n(x)| : x \in S\} = 0$.

pf.

Let $\epsilon > 0$. Since $\sum g_n$ converges uniformly, $\exists N > 0$ such that, $\exists \forall x \in S$,

$$n \geqslant m > N \text{ implies } \left|\sum_{k=m}^{n} g_k(x)\right| < \epsilon.$$

If we set $n = m$, then, $\exists \forall x \in S$,

$$\begin{aligned} & n > N \text{ implies } |g_n(x)| < \epsilon, \\ \Longrightarrow\ & n > N \text{ implies } \sup\{|g_n(x)| : x \in S\} < \epsilon, \\ \Longleftrightarrow\ & \lim_{n\to\infty} \sup\{|g_n(x)| : x \in S\} = 0. \end{aligned}$$

■

§26. Diff. & Int. of Power Series

(In this page, $\sum_{n=0}^{\infty} a_n x^n$ is a power series with the radius of convergence R.)

· Continuity of Power Series

- For $0 < R_1 < R$, $\sum a_n x^n$ is continuous on $[-R_1, R_1]$.

pf.

Since $|a_n x^n| \leqslant |a_n| R_1^{\,n}$, where $\sum |a_n| R_1^{\,n} < \infty$, by the Weierstrass M-test, $\sum a_n x^n$ converges uniformly on $[-R_1, R_1]$.
And also each $a_n x^n$ is continuous on $[-R_1, R_1]$, thus $\sum a_n x^n$ converges uniformly on $[-R_1, R_1]$ to a continuous function. ■

- $\sum a_n x^n$ is continuous on $(-R, R)$.

pf.

For $x_0 \in (-R, R)$, $\exists R_1$ s.t. $x_0 \in (-R_1, R_1)$.
Thus $\sum a_n x^n$ is cont. at any $x_0 \in (-R, R)$, which is equivalent to $\sum a_n x^n$ is cont. on $(-R, R)$. ■

· Diff. & Int. of Power Series

- **Radius of Convergence**

: $\sum_{n=1}^{\infty} n a_n x^{n-1}$ and $\sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ also have radius of convergence R.

pf.

Since $\beta := 1/R = \limsup |a_n|^{1/n}$,

(i) $\limsup |n a_n|^{1/n} = \limsup n^{1/n} |a_n|^{1/n} = \beta$;

(ii) $\limsup \left|\frac{a_n}{n+1}\right|^{1/n} = \limsup \frac{|a_n|^{1/n}}{|n+1|^{1/n}} = \beta$,

which implies the proposition stands. ■

- **Integration of Power Series**

: $\int_0^x \left(\sum_{n=0}^{\infty} a_n t^n\right) \mathrm{d}t = \sum_{n=0}^{\infty} \int_0^x a_n t^n \, \mathrm{d}t = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$.
($\Longleftrightarrow$ Power series can be integrated term by term.)

pf.

Let $f(x) = \sum a_n x^n$, and fix $x \in (-R, 0)$.
Then, $\sum a_k t^k$ converges uniformly on $[x, 0]$.
Thus, by Thm 25.2, we have

$$\begin{aligned}\int_x^0 \left(\sum_{n=0}^{\infty} a_n t^n\right) \mathrm{d}t &= \lim_{n\to\infty} \int_x^0 \left(\sum_{k=0}^{n} a_k x^k\right) \mathrm{d}t \\ &= \lim_{n\to\infty} \sum_{k=0}^{n} \int_x^0 a_k t^k \, \mathrm{d}t \\ &= \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}.\end{aligned}$$

■

- **Differentiation of Power Series**

: For $f(x) = \sum_{n=0}^{\infty} a_n x^n$,

(i) f is d.able on $(-R, R)$;

(ii) $f'(x) = \sum_{n=0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}x} a_n x^n = \sum_{n=1}^{\infty} n a_n x^{n-1}$.

pf.

Let $g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$, and observe that its radius of convergence is R.
By Thm 26.4, for $|x| < R$,

$$\begin{aligned}\int_0^x g(t) \, \mathrm{d}t &= \sum_{n=1}^{\infty} \int_0^x n a_n t^{n-1} \, \mathrm{d}t \\ &= \sum_{n=1}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^n - a_0.\end{aligned}$$

Thus, for $0 < R_1 < R$ and $|x| \leqslant R_1$,

$$\begin{aligned}f(x) &= \int_0^x g(t) \, \mathrm{d}t + a_0 \\ &= \int_{-R_1}^x g(t) \, \mathrm{d}t \underbrace{- \int_{-R_1}^0 g(t) \, \mathrm{d}t + a_0}_{=:\, k} \\ &= \int_{-R_1}^x g(t) \, \mathrm{d}t + k.\end{aligned}$$

By above and since g is continuous on $(-R, R)$, by Thm 34.4(FTC2),
f is d.able and $f'(x) = g(x)$ on $(-R, R)$. ■

· Abel's Theorem

- If $\sum a_n x^n$ converges at $x = R$ or $x = -R$, then f is cont. at $x = R$ or $x = -R$, respectively.

§-. Midterm Test Review

· Continuity of Functions

: f is continuous at x_0, if given $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$|x - x_0| < \delta \ \wedge \ x \in \operatorname{dom} f \implies |f(x) - f(x_0)| < \epsilon.$$

· Convergence of Sequences

: For a sequence (a_n), $\lim a_n = a$ if
given $\epsilon > 0$, $\exists N > 0$ s.t.

$$n > N \quad \text{implies} \quad |a_n - a| < \epsilon.$$

· Convergence of Sequences of Functions

- For a sequence (f_n) of ftns on $S \subseteq \mathbb{R}$,
$\lim f_n(x) = f(x)$ if given $\epsilon > 0$ and $x \in S$, $\exists N > 0$ s.t.

$$n > N \quad \text{implies} \quad |f_n(x) - f(x)| < \epsilon.$$

- $f_n \longrightarrow f$ uniformly on S if given $\epsilon > 0$, $\exists N > 0$ s.t.

$$n > N \ \wedge \ x \in S \quad \text{implies} \quad |f_n(x) - f(x)| < \epsilon.$$

§28. Basic Properties of the Derivative

· Derivative

- Definition

: f is differentiable at a, or f has a derivative at a, if the limit $f'(a) = \lim_{x\to a} \dfrac{f(x)-f(a)}{x-a}$ exists, and is finite.

(f: real-valued function on an open interval I, $a \in I$.)

- Basic Derivatives

· $f(x) = x^n$; $f'(a) = na^{n-1}$. $(n \in \mathbb{N},\ a \in \operatorname{dom} f)$

- Continuity of Diff.ble Function

: If f is diff.ble at a, then f is cont. at a.

pf.

Since f is diff.ble at a, there exists

$$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$$

s.t. $|f'(a)| < +\infty$.

We derive that for $x \neq a$,

$$f(x) = \frac{f(x)-f(a)}{x-a}\cdot(x-a) + f(a).$$

Take the limit "$\lim_{x\to a}$":

$$\begin{aligned}\lim_{x\to a} f(x) &= \lim_{x\to a}\left(\frac{f(x)-f(a)}{x-a}\cdot(x-a)+f(a)\right)\\ &= f'(a)\cdot 0 + f(a) = f(a),\end{aligned}$$

so we conclude that f is continuous on a. ■

· Differential Rules

: For diff.ble ftns f & g at a and $c \in \mathbb{R}$,
cf, $f+g$, fg, f/g are also diff.ble at a;

- $(cf)'(a) = c\cdot f'(a)$,
- $(f+g)'(a) = f'(a) + g'(a)$,
- $(fg)(a) = f'g(a) + fg'(a)$, *[product rule]*
- $\left(\dfrac{f}{g}\right)'(a) = \dfrac{f'(a)g(a) - f(a)g'(a)}{(g(a))^2}$. *[quotient rule]*

(For only on the quotient rule, a must satisfy $g(a) \neq 0$.)

pf. (Constant Multi. & Addition)

(i) $$\begin{aligned}(cf)'(a) &= \lim_{x\to a}\frac{(cf)(x)-(cf)(a)}{x-a}\\ &= c\lim_{x\to a}\frac{f(x)-f(a)}{x-a} = cf'(a).\end{aligned}$$

(ii) $$\begin{aligned}(f+g)'(a) &= \lim_{x\to a}\frac{(f+g)(x)-(f+g)(a)}{x-a}\\ &= \frac{f(x)-f(a)+g(x)-g(a)}{x-a}\\ &= f'(a)+g'(a).\end{aligned}$$ ■

pf. (Product & Quotient)

(iii) $$\begin{aligned}(fg)'(a) &= \lim_{x\to a}\frac{(fg)(x)-(fg)(a)}{x-a}\\ &= \lim_{x\to a}\frac{f(x)g(x)-f(a)g(a)}{x-a}\\ &= \lim_{x\to a}\frac{(fg)(x)-f(x)g(a)+f(x)g(a)-(fg)(a)}{x-a}\\ &= f'(a)g(a)+f(a)g'(a).\end{aligned}$$

(iv) Let $I \subseteq \operatorname{dom} f$ be an open interval including a s.t. $g(x) \neq 0$ on $x \in I$. ($\exists$ such I by the cont. at a of g.)
Let $x \in I \setminus \{a\}$. Then

$$\begin{aligned}&\left(\frac{f}{g}\right)(x) - \left(\frac{f}{g}\right)(a)\\ &= \frac{f(x)g(a)-f(a)g(x)}{g(x)g(a)}\\ &= \frac{f(x)g(a)-(fg)(a)+(fg)(a)-f(a)g(x)}{g(x)g(a)}\\ &= \frac{g(a)(f(x)-f(a))-f(a)(g(x)-g(a))}{g(x)g(a)},\end{aligned}$$

so

$$\begin{aligned}&\frac{(f/g)(x)-(f/g)(a)}{x-a}\\ &= \frac{g(a)\dfrac{f(x)-f(a)}{x-a} - f(a)\dfrac{g(x)-g(a)}{x-a}}{g(x)g(a)}\\ &\longrightarrow \frac{g(a)f'(a)-f(a)g'(a)}{(g(a))^2}\end{aligned}$$

as $x \longrightarrow a$. ■

· Chain Rule

: If f is diff.ble at a and so is g at $f(a)$,
then $g\circ f$ is diff.ble at a, and $(g\circ f)'(x) = g'(f(x))f'(x)$.

pf.

$$\begin{aligned}(g\circ f)'(a) &= \lim_{x\to a}\frac{(g\circ f)(x)-(g\circ f)(a)}{x-a}\\ &= \lim_{x\to a}\frac{g(f(x))-g(f(a))}{x-a}\\ &= \lim_{x\to a}\left(\frac{g(f(x))-g(f(a))}{f(x)-f(a)}\cdot\frac{f(x)-f(a)}{x-a}\right)\\ &= g'(f(a))f'(a).\end{aligned}$$ ■

§29. The Mean Value Theorem

· Derivative at Max/Min Point

: For a function f on open inteval I and $x_0 \in I$,

if $\begin{cases} f \text{ has max/min value at } x_0, \\ f \text{ is differentiable at } x_0, \end{cases}$

then $f'(x_0) = 0$.

pf.

Let f be a ftn on $I = (a, b)$ max. at x_0. ····· (1)
Assume that $f'(x_0) > 0$. Since

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0},$$

$\exists \delta > 0$ s.t. $(x_0 - \delta, x_0 + \delta) \subseteq (a, b)$ and
$0 < |x - x_0| < \delta$ implies $\dfrac{f(x) - f(x_0)}{x - x_0} > 0$.
But if we select $x \in (x_0, x_0 + \delta)$, then $x - x_0 > 0$,
thus $f(x) - f(x_0) > 0$, which contrd. to (1).
Hence $f'(x_0) \not> 0$, $\therefore$ $f'(x_0) \leqslant 0$. ····· (2)
By similar way, we can show $f'(x_0) \geqslant 0$. ····· (3)
By (2) & (3), we have $f'(x_0) = 0$. ■

· Rolle's Theorem

: For a function f on $[a, b]$, if

$\begin{cases} f \text{ is continuous on } [a, b], \\ f \text{ is differentiable on } (a, b), \\ f(a) = f(b), \end{cases}$

then $\exists$ (at least one) $x \in (a, b)$ s.t. $f'(x) = 0$.

pf.

Since f is cont. on $[a, b]$, $\exists x_0, y_0 \in [a, b]$ s.t.
$f(x_0) \leqslant f(x) \leqslant f(y_0)$ for all $x \in [a, b]$, ····· (1)
i.e., f has its min. value at x_0, and max. value at y_0.

(i) If x_0 and y_0 are both endpoints of $[a, b]$,
then $f(a) = f(b) = f(x_0) = f(y_0)$,
but by (1), f is constant ftn, thus $f'(x) = 0$ for all $x \in [a, b]$.

(ii) If x_0 or y_0 belongs to (a, b),
then $\exists x \in (a, b)$ s.t. $f(x)$ is max. or min. value,
which implies $f'(x) = 0$ by the theorem ubove. ■

· Mean Value Theorem

: For a function f on $[a, b]$, if

$\begin{cases} f \text{ is continuous on } [a, b], \\ f \text{ is differentiable on } (a, b), \end{cases}$

then $\exists$ (at least one) $x \in (a, b)$ s.t. $f'(x) = \dfrac{f(b) - f(a)}{b - a}$.

pf.

Let $L(x) = f(x) - \dfrac{f(b) - f(a)}{b - a}(x - a) + f(a)$.
Then L is cont. on $[a, b]$ and diff.ble on (a, b), and since

$$L(a) = L(b) = 0,$$

by the Rolle's theorem, $\exists x \in (a, b)$ s.t. $L'(x) = 0$, thus
$f'(x) = \dfrac{f(b) - f(a)}{b - a}$. ■

· Corollaries of MVT

- For a diff.ble function f on an open interval I,
  if $f'(x) = 0$ for all $x \in I$, then f is constant ftn. on I.

  pf.

  Assume that f is not const. ftn. on I.
  Then $\exists x_1, x_2 \in I$ s.t. $x_1 \neq x_2$ and $f(x_1) \neq f(x_2)$.
  Since f is cont. on $[x_1, x_2]$ and diff.ble on (x_1, x_2),
  by the Mean Value Theorem, $\exists x \in (x_1, x_2)$ s.t.

  $$f'(x) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \neq 0,$$

  which contrd. to the assumption.
  Thus $f(x_1) = f(x_2)$. ■

- If $f' = g'$ on an open interval I,
  then $\exists c \in \mathbb{R}$ s.t. $f(x) = g(x) + c$ for all $x \in I$.

  pf.

  Apply the corollary above to $h = f - g$. ■

· Inc./Dec. of Functions

- Definition

: For real-valued ftn. f on an open interval I,
and $x_1, x_2 \in I$ s.t. $x_1 < x_2$,

- f is strictly increasing if $f(x_1) < f(x_2)$.
- f is strictly decreasing if $f(x_1) > f(x_2)$.
- f is increasing if $f(x_1) \leqslant f(x_2)$.
- f is decreasing if $f(x_1) \geqslant f(x_2)$.

- Property

: For a diff.ble f on an open interval I,

- f is strictly increasing if $f'(x) > 0$ for all $x \in I$.
- f is strictly decreasing if $f'(x) < 0$ for all $x \in I$.
- f is increasing if $f'(x) \geqslant 0$ for all $x \in I$.
- f is decreasing if $f'(x) \leqslant 0$ for all $x \in I$.

· Intermediate Value Theorem for Derivatives

: For a diff.ble function f on $I = (a, b)$,
let $x_1, x_2 \in (a, b)$ s.t. $x_1 < x_2$,
and $c \in \mathbb{R}$ lies between $f'(x_1) < f'(x_2)$.

Then $\exists$ at least one $x \in (x_1, x_2)$ s.t. $f'(x) = c$.

pf.

Assume $f'(x_1) < f'(x_2)$, and define $g(x) = f(x) - cx$.

Then $g'(x) = f'(x) - c$, hence $g'(x_1) < 0 < g'(x_2))$,
and g is cont. on $[x_1, x_2]$, so g is bounded on $[x_1, x_2]$.

Assume g has its minimum value at $x_0 \in [x_1, x_2]$.
Then, since

$$g'(x_1) = \lim_{y \to x_1^+} \frac{g(y) - g(x_1)}{y - x_1} < 0,$$

$g(y) - g(x_1) < 0$ for $y_1 = x_1 + \epsilon$ for some $\epsilon > 0$ s.t. $y_1 < x_2$.
Then $g(y_1) < g(x_1)$, thus $\min_{x \in I} g(x) \neq g(x_1)$. Similarly,
$\exists y_2 \in (x_1, x_2)$ s.t. $g(y_2) < g(x_2)$, thus $\min_{x \in I} g(x) \neq g(x_2)$.
Therefore, $x_0 \notin \{x_1, x_2\}$, hence x_0 s.t. $\min_{x \in I} g(x) = g(x_0)$
belongs to (x_1, x_2), thus $g'(x_0) = 0$, $f'(x_0) = c$. ■

· Derivatives of Inverse Functions

: For one-to-one continuous ftn. f on an open interval I,
if f is diff.ble on $x_0 \in I$ and if $f'(x_0) \neq 0$, then
f^{-1} is diff.ble at $y_0 = f(x_0)$, and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)} \left(= \frac{1}{f'(f^{-1}(y_0))} \right).$$

pf.

Note that $J = f(I)$ is also an open interval.
Since $f'(x_0) \neq 0$, $f(x) \neq f(x_0)$ for x near x_0, thus

$$\lim_{x \to x_0} \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{f'(x_0)}.$$

Then, for given $\epsilon > 0$, $\exists \delta > 0$ s.t. $0 < |x - x_0| < \delta$ implies

$$\left| \frac{x - x_0}{f(x) - f(x_0)} - \frac{1}{f'(x_0)} \right| = \left| \frac{g(y) - g(y_0)}{y - y_0} - \frac{1}{f'(x_0)} \right| < \epsilon$$

with $y = f(x)$ on I, $y_0 = f(x_0)$, and $g = f^{-1}$.
Since g is cont. at y_0, $\exists \eta > 0$ s.t. $0 < |y - y_0| < \eta$ implies

$$|g(y) - g(y_0)| = |g(y) - x_0| < \delta.$$

Hence we obtain

$$0 < |y - y_0| < \eta \quad \text{implies} \quad \left| \frac{g(y) - g(y_0)}{y - y_0} - \frac{1}{f'(x_0)} \right| < \epsilon,$$

which is equivalent to

$$\lim_{y \to y_0} \frac{g(y) - g(y_0)}{y - y_0} = (f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

■

§31. Taylor's Theorem

· Taylor Series

- Coefficients of Power Series

: Consider a power series $f(x)=\sum_{k=0}^{\infty}a_kx^k$ for $|x|<R$. Then f is indefinitely diff.ble on $(-R,\,R)$;

· $f'(x)=\sum_{k=1}^{\infty}ka_kx^{k-1}$,

· $f''(x)=\sum_{k=2}^{\infty}k(k-1)a_kx^{k-2}$,

$\vdots$

· $f^{(n)}(x)=\sum_{k=n}^{\infty}k(k-1)\cdots(k-n+1)a_kx^{k-n}$,

thus $f^{(n)}(0)=n\times(n-1)\times\cdots2\times1\times a_n=n!\,a_n$,

so $a_n=\dfrac{f^{(n)}(0)}{n!}$, hence we have

$$f(x)=\sum_{k=0}^{\infty}\frac{f^{(k)}(0)}{k!}x^k\quad\text{for }|x|<R.$$

- Taylor Series

: For a ftn. f possessing derivatives of all orders at $c\in\mathrm{dom}(f)$, then the Taylor series for f about c is

$$\sum_{k=0}^{\infty}\frac{f^{(k)}(c)}{n!}(x-c)^k.$$

- Remainder

: For $n\geqslant1$, the remainder is defined by

$$R_n(x)=f(x)-\sum_{k=0}^{n-1}\frac{f^{(k)}(c)}{n!}(x-c)^k.$$

- $\lim R_n(x)=0$ if and only if $f(x)=\sum_{k=0}^{\infty}\dfrac{f^{(k)}(c)}{n!}(x-c)^k$.

- Example of Taylor Series

· $e^x=\sum_{k=0}^{\infty}\dfrac{x^k}{k!}$.

· $\dfrac{1}{1-x}=\sum_{k=0}^{\infty}x^k\implies\ln(1-x)=\sum_{k=1}^{\infty}\dfrac{x^k}{k}.\quad(|x|<1)$

· $\sin x=\sum_{k=0}^{\infty}\dfrac{(-1)^k}{(2k+1)!}x^{2k+1}$.

· $\cos x=\sum_{k=0}^{\infty}\dfrac{(-1)^{k+1}}{(2k)!}x^{2k}$.

· Taylor's Theorem

: For a ftn. f on $(a,\,b)$, let $c\in(a,\,b)$, and assume $\exists f^{(n)}$ on $(a,\,b)$. Then for each $x\neq c$ in $(a,\,b)$, $\exists y$ between x and c s.t. $R_n(x)=\dfrac{f^{(n)}(y)}{n!}(x-c)^n$.

This form is called *Lagrange form of the remainder*.

pf.

Let $x\neq c$, $n\geqslant1$ and M be the unique solution of the eq.

$$f(x)=\sum_{k=0}^{n-1}\frac{f^{(k)}(c)}{k!}(x-c)^k+\frac{M}{n!}(x-c)^n.\qquad(1)$$

We have to show $R_n(x)=M(x-c)^n/n!=f^{(n)}(y)(x-c)^n/n!$, i.e., $M=f^{(n)}(y)$ for some y between x and c.

Let $g(t)=\sum_{k=0}^{n-1}\dfrac{f^{(k)}(c)}{k!}(t-c)^k+\dfrac{M}{n!}(t-c)^n-f(t)$.

Then $g(c)=f(c)-f(c)=0$, $g'(c)=f'(c)-f'(c)=0$, …, $g^{(n-1)}(c)=f^{(n-1)}(c)-f^{(n-1)}(c)=0$.

By the choice of M on (1), $g(x)=0$. Since g is cont. & diff.ble on $(x,\,c)$ or $(c,\,x)$, by the Rolle's theorem, $\exists x_1$ between x and c s.t. $g'(x_1)=0$.

Then $g'(x_1)=g'(c)=0$, thus $\exists x_2$ bt. x_1 & c s.t. $g''(x_2)=0$.

$\vdots$

$g^{(n-1)}(x_{n-1})=g^{(n-1)}(c)=0$, $\exists x_n$ between x_{n-1} & c s.t. $g^{(n)}(x_n)=0$.

So $g^{(n)}(x_n)=n!\cdot\dfrac{M}{n!}(x-c)^0-f^{(n)}(x_n)=M-f^{(n)}(x_n)=0$, we derive $M=f^{(n)}(y)$ with $y=x_n$.

Hence we have $R_n(x)=\dfrac{f^{(n)}(y)}{n!}(x-c)^n$. ■

- Corollary

: For a function f on $(a,\,b)$, if all $f^{(n)}$ exists on $(a,\,b)$, and if f is bounded by a single constant C, then

$$\lim R_n(x)=0\quad\text{for all }x\in(a,\,b).$$

pf.

By the Taylor's theorem, $\exists y$ between x and $c\in(a,\,b)$ s.t.

$$|R_n(x)|=\left|\frac{f^{(n)}(y)}{n!}(x-c)^n\right|=\frac{|f^{(n)}(y)|}{n!}|x-c|^n\leqslant\frac{C}{n!}|x-c|^n.$$

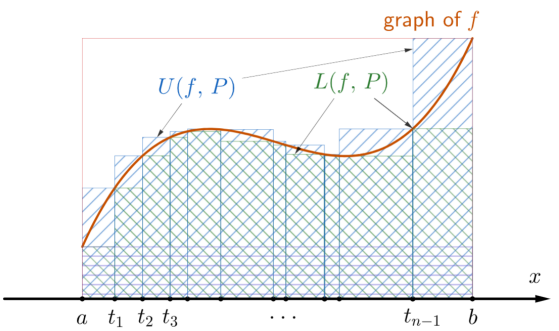
Since $|x-c|^n/n!\longrightarrow0$ as $n\to\infty$, by the squeeze theorem, we have $\lim R_n(x)=0$. ■

§32. The Riemann Integral

· Darboux Integrals

: Let f be a bounded function on $[a, b]$, and $S \subseteq [a, b]$.

- **Darboux Sum**
  - $M(f, S) = \sup\{f(x) : x \in S\}$.
  - $m(f, S) = \inf\{f(x) : x \in S\}$.
  - **Partition of** $[a, b]$
    : $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$.
  - **Upper Darboux Sum of** f **with respect to** P
    : $U(f, P) \equiv \sum_{k=1}^{n} M(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1})$.
  - **Lower Darboux Sum of** f **with respect to** P
    : $L(f, P) \equiv \sum_{k=1}^{n} m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1})$.
  - $m(f, [a, b])(b-a) \leqslant L(f, P) \leqslant U(f, P) \leqslant M(f, [a, b])(b-a)$
    for any partition P of $[a, b]$.



  - A partition Q is finer than P, if $|P| \geqslant |Q|$.
    Especially, if $P \subseteq Q$, Q is called a refinement of P.
    As P gets finer, $U(f, P) \searrow$ & $L(f, P) \nearrow$.

- **Darboux Integral**
  - **Upper Darboux Integral** (often denoted by $\overline{\int_a^b} f$)
    : $U(f) \equiv \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}$.
  - **Lower Darboux Integral** (often denoted by $\underline{\int_a^b} f$)
    : $L(f) \equiv \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}$.
  - If f is bounded on $[a, b]$, then $L(f) \leqslant U(f)$.
  - **Darboux Integral**
    : If $U(f) = L(f)$, then f is integrable on $[a, b]$,
    and its Darboux integral on $[a, b]$ is defined by

$$\left(\mathcal{D}\int_a^b f = \right) \int_a^b f = \int_a^b f(x)\,dx \equiv U(f) = L(f).$$

- **Theorems** (f: a bdd. ftn. on $[a, b]$.)
  - For partitions P & Q of $[a, b]$ s.t. $P \subseteq Q$,

$$L(f, P) \leqslant L(f, Q) \leqslant U(f, Q) \leqslant U(f, P).$$

pf.: $L(f, P) \leqslant L(f, Q)$

Without lost of generallity, for a partition $P = \{a = t_0 < \cdots < t_n = b\}$, it suffices to show where $Q = P \cup \{u\}$ and $u \in (t_{k-1}, t_k)$ f.s. k.

Let $m(f, [p, q])(q-p) = \mu_{p,q}(q-p) = \Delta_{p,q}$ for $p, q \in Q$ s.t. $p < q$. Then

$$\begin{aligned} L(f, Q) - L(f, P) &= \Delta_{t_{k-1},u} + \Delta_{u,t_k} - \Delta_{t_{k+1},t_k} \\ &= \Delta_{t_{k-1},u} + \Delta_{u,t_k} \\ &\quad - \mu_{t_{k+1},t_k}(t_k - u + u - t_{k-1}) \\ &\geqslant 0, \end{aligned}$$

since $\mu_{t_{k-1},t_k} \leqslant \min\{\mu_{t_{k-1},u}, \mu_{u,t_k}\}$,
thus we have $L(f, Q) \geqslant L(f, P)$. ■

  - For partitions P & Q of $[a, b]$,

$$L(f, P) \leqslant U(f, Q).$$

pf.

Since $P \cap Q$ is a partition of $[a, b]$,
and $P, Q \subseteq P \cap Q$, we have

$$\begin{aligned} L(f, P) &\leqslant L(f, P \cap Q) \\ &\leqslant U(f, P \cap Q) \leqslant U(f, Q), \end{aligned}$$

hence $L(f, P) \leqslant U(f, Q)$. ■

  - $L(f) \leqslant U(f)$.

pf.

Let P be a partition of $[a, b]$.
Since $L(f, P) \leqslant U(f, Q)$ for any partition Q of $[a, b]$, $L(f, P) \leqslant U(f)$.
Thus we have $L(f) \leqslant U(f)$. ■

  - **Cauchy Criterion for Integrability (1)**
    : f is integrable if and only if
    for each $\epsilon > 0$, $\exists$ a partition P of $[a, b]$ s.t.

$$U(f, P) - L(f, P) < \epsilon.$$

pf. ($\Longleftarrow$)

For $\epsilon > 0$ and some partition P,

$$\begin{aligned} U(f) \leqslant U(f, P) &= U(f, P) - L(f, P) + L(f, P) \\ &< \epsilon + L(f, P) \leqslant \epsilon + L(f). \end{aligned}$$

Since ϵ is arbitrary, we have $U(f) \leqslant L(f)$.
But it is trivial that $U(f) \geqslant L(f)$, thus we have $U(f) = L(f)$, which implies f is int.ble. □

pf. ($\Longrightarrow$)

Let $\epsilon > 0$. As P gets finer, i.e., $|P| \longrightarrow \infty$, $(L(f, P), U(f, P)) \longrightarrow (L(f), U(f))$, thus $\exists P_1$ & P_2: partitions of $[a, b]$ s.t.

$$L(f, P_1) > L(f) - \epsilon/2,\ U(f, P_2) < U(f) + \epsilon/2.$$

Then, for $P = P_1 \cup P_2$,

$$\begin{aligned} U(f, P) - L(f, P) &\leqslant U(f, P_2) - L(f, P_1) \\ &= U(f) - L(f) + \epsilon = \epsilon. \end{aligned}$$

■

· Suppose there exist $\begin{cases} (U_n): \text{seq. of upper D. sum,} \\ (L_n): \text{seq. of lower D. sum.} \end{cases}$

If $\lim (U_n - L_n) = 0$, then f is integrable, and

$$\int_a^b f = \lim L_n = \lim U_n.$$

pf.

Integrability. Let $U_n = U(f, P_n)$, $L_n = (f, Q_n)$ for seq. (P_n), (Q_n) of pttn., and $P = P_n \cup Q_n$.

We derive

$$U(f, P) - L(f, P) \leqslant U(f, P_n) - L(f, Q_n) \quad (1)$$

since P is finer than both P_n and Q_n, and (1) converges to 0 by the assumption.

Hence, for any $\epsilon > 0$, $\exists N > 0$ s.t.

$$n > N \quad \text{implies} \quad U(f, P) - L(f, P) < \epsilon.$$

Therefore, with P for such n, we can show f is integrable on $[a, b]$. □

. . . . . .

Equality. Note that

$$L_n \leqslant L(f) = \int_a^b f = U(f) \leqslant U_n$$

since f is integrable by the proof above.

Then we have

$$\begin{cases} 0 \leqslant \int_a^b f - L_n \leqslant U_n - L_n, \\ L_n - U_n \leqslant \int_a^b -U_n \leqslant 0, \end{cases}$$

but if we take the limit $n \longrightarrow \infty$,

$$\begin{cases} 0 \leqslant \int_a^b f - \lim L_n \leqslant \lim (U_n - L_n), \\ 0 = \lim (L_n - U_n) \leqslant \int_a^b - \lim U_n \leqslant 0, \end{cases}$$

hence we conclude $\int_a^b f = \lim L_n = \lim U_n$. ■

· Riemann Integrals

- Mesh

: The mesh of $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ is defined by a number s.t.

$$\operatorname{mesh}(P) = \max \{t_k - t_{k-1} : k = 1, 2, \ldots, n\}.$$

- Cauchy Criterion for Integrability (2)

: A bdd. ftn f on $[a, b]$ is integrable if and only if for each $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$\operatorname{mesh}(P) < \delta \quad \text{implies} \quad U(f, P) - L(f, P) < \epsilon.$$

pf. ($\Longrightarrow$)

Let $\epsilon > 0$, and select a partition

$$P_0 = \{a = u_0 < u_1 < \cdots < u_m = b\}$$

s.t. $U(f, P_0) - L(f, P_0) < \epsilon/2$. ····· (1)

(Its existence is proved by Thm 32.5.)

Since f is bdd. on $[a, b]$, $\exists B \geqslant 0$ s.t.

$$-B \leqslant f(x) \leqslant B \quad \text{and} \quad -B \leqslant -f(x) \leqslant B.$$

for all $x \in [a, b]$.

Let $\delta = \epsilon/(8B(m-1))$, and consider a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ s.t. $\operatorname{mesh}(P) < \delta$, and let Q be a partition s.t. $Q = P_0 \cup P$.

(i) If $|Q| = |P| + 1$, i.e., $Q \setminus P = \{u\}$ for some $u \in (t_{k-1}, t_k)$,

$$\begin{aligned} L(f, Q) - L(f, P) &= \Delta_{t_{k-1}, u} + \Delta_{u, t_k} - \Delta_{t_{k-1}, t_k} \\ &\leqslant 2B(t_k - t_{k-1}) \\ &< 2B\delta. \end{aligned}$$

(ii) If $|Q| - |P| \leqslant 1$, then $|Q| \leqslant |P| + (m-1)$, by (i) and the inductive argument, we have

$$L(f, Q) - L(f, P) < 2B(m-1)\delta = \epsilon/4. \quad \cdots\cdots (2)$$

Since $P_0 \subseteq Q$, $L(f, P_0) \leqslant L(f, Q)$, thus $L(f, P_0) - L(f, P) \leqslant L(f, Q) - L(f, P) < \epsilon/4$. ····· (3)

Similarly, we can derive

$$\min \{U(f, P) - U(f, Q), U(f, P) - U(f, P_0)\} < \epsilon/4. \quad \cdots\cdots (4)$$

By (1)-(4), we conclude

$$\begin{aligned} U(f, P) - L(f, P) &< U(f, P_0) + \epsilon/4 - L(f, P_0) + \epsilon/4 \\ &= U(f, P_0) - L(f, P_0) + \epsilon/2 \\ &< \epsilon/2 + \epsilon/2 = \epsilon. \end{aligned}$$

■

- **Riemann Sum**
  : A Riemann sum of f associated with P is defined by

$$(S=)\sum_{k=1}^{n} f(x_k)(t_k - t_{k-1})$$

  with a bdd. ftn. f on $[a,\, b]$,
  a partition $P = \{a = t_0 < \cdots < t_n = b\}$ of $[a,\, b]$, and
  $x_k \in [t_{k-1},\, t_k]$ for $k = 1,\, 2,\, \ldots,\, n$.

  · **Right/Left Riemann Sum**
    : A Riemann sum where $x_k = t_k$(R) / t_{k-1}(L).

  · Since the choice of x_n is arbitrary, $|(\mathrm{R}.\Sigma)| = \infty$.

  · **Inequality between D.Sum & R.Sum**
    : For any R.Sum S with P, $L(f,\, P) \leqslant S \leqslant U(f,\, P)$.

- **Riemann Integrability**
  : f is Riemann integrable on $[a,\, b]$,
  if $\exists r \in \mathbb{R}$ s.t. for each $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$\mathrm{mesh}(P) < \delta \quad \text{implies} \quad |S - r| < \epsilon$$

  for any Riemann sum S of f with P.

- **Riemann Integral**
  : The Riemann integral of f on $[a,\, b]$ is defined by
  such $r \in \mathbb{R}$ (s.t. $|S-r| < \epsilon$) on def. of Riemann int.bility,
  and we denote it

$$r = \mathcal{R}\int_a^b f.$$

- A function f is Riemann int.ble on $[a,\, b]$,
  if and only if it is [Darboux] int.ble on $[a,\, b]$.

pf. ($\Longleftarrow$)

Suppose f is D. int.ble on $[a,\, b]$.
Then, for $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$\mathrm{mesh}(P) < \delta \quad \text{implies} \quad U(f,\, P) - L(f,\, P) < \epsilon$$

for any partition P of $[a,\, b]$.
Let P be a partition of $[a,\, b]$ s.t. $\mathrm{mesh}(P) < \delta$. Then

$$U(f,\, P) < L(f,\, P) + \epsilon \leqslant L(f) + \epsilon = \int_a^b f + \epsilon, \quad (1)$$

$$L(f,\, P) > U(f,\, P) - \epsilon \geqslant U(f) - \epsilon - \int_a^b f - \epsilon. \quad (2)$$

For any Riemann sum S with partition P,

$$L(f,\, P) \leqslant S \leqslant U(f,\, P),$$

thus, by (1) & (2),

$$\int_a^b f - \epsilon < S < \int_a^b f + \epsilon \iff \left| S - \int_a^b f \right| > \epsilon,$$

hence f is Riemann int.ble, and $\mathcal{R}\int_a^b f = \int_a^b f$. □

pf. ($\Longrightarrow$)

Conversely, suppose f is R. int.ble on $[a,\, b]$.
Then, for $\epsilon > 0$, $\exists d$, $r > 0$ s.t.

$$\mathrm{mesh}(P) < \delta \quad \text{implies} \quad |S - r| < \epsilon$$

for any Riemann sum S with partition P.
Consider a partition $P = \{a = t_0 < \cdots < t_n = b\}$
s.t. $\mathrm{mesh}(P) < \delta$.
Let $x_k \in [t_{k-1},\, t_k]$ s.t. $f(x_k) < m(f,\, [t_{k-1},\, t_k]) + \epsilon$
for $k = 1,\, 2,\, \ldots,\, n$.
Then we have

$$\sum_{k=1}^{n} f(x_k)(t_k - t_{k-1}) < \sum_{k=1}^{n} m(f,\, [t_{k-1},\, t_k])(t_k - t_{k-1}) + \sum_{k=1}^{n} \epsilon(t_k - t_{k-1}),$$

hence $S < L(f,\, P) + \epsilon(b-a)$.
It follows that

$$L(f) \leqslant L(f,\, P) > S - \epsilon(b-a) > r - \epsilon(b-a+1) \geqslant r$$

since ϵ is arbitrary.
Similarly, we can show $U(f) \leqslant r$.
Since $L(f) \leqslant U(f)$, we have $L(f) = U(f) = r$,
hence f is [D.] integrable, and $\int_a^b f = \mathcal{R}\int_a^b f$. ■

- For bounded and Riemann int.ble ftn. f on $[a,\, b]$,
  let (S_n) be a seq. of Riemann sum with
  a partition seq. (P_n) s.t. $\lim \mathrm{mesh}(P_n) = 0$.
  Then we have

$$\int_a^b f = \lim S_n.$$

pf.

Let $\epsilon > 0$. By the assumption, $\exists N > 0$ s.t.

$$n > N \quad \text{implies} \quad \mathrm{mesh}(P_n) < \delta$$

for $\delta > 0$ s.t.

$$\mathrm{mesh}(P) < \delta \quad \text{implies} \quad \left| S - \int_a^b f \right| < \epsilon$$

for any Riemann sum S with any P.
(Such δ exists since f is R. int.ble!)
Then, for $n > N$, $\left| S_n - \int_a^b f \right| < \epsilon$,
hence we conclude $\lim S_n = \int_a^b f$. ■

§33. Properties of the Riemann Integral

· Integrability

- Every monotonic ftn. f on $[a, b]$ is integrable.

pf. (Increasing)

Assume f is inc. ftn. on $[a, b]$ and $f(a) < f(b)$.
Then $f(x) \in [f(a), f(b)]$ for any $x \in [a, b]$,
hence f is bounded on $[a, b]$.
Let $\epsilon > 0$, $P = \{a = t_0 < \cdots < t_n = b\}$ with

$$\mathrm{mesh}(P) < \frac{\epsilon}{f(b) - f(a)}.$$

Then we derive, for $\Delta_k = t_k - t_{k-1}$ and $I_k = [t_{k-1}, t_k]$,

$$\begin{aligned} &U(f, P) - L(f, P) \\ &= \sum_{k=1}^{n} [M(f, I_k) - m(f, I_k)]\Delta_k \\ &< \sum_{k=1}^{n} [f(t_k) - f(t_{k-1})] \cdot \frac{\epsilon}{f(b) - f(a)} \\ &= [f(b) - f(a)] \cdot \frac{\epsilon}{f(b) - f(a)} = \epsilon, \end{aligned}$$

hence f is integrable on $[a, b]$. ■

- Every continuous ftn. on $[a, b]$ is integrable.

pf.

By the assumption, f is bounded on $[a, b]$,
thus f is uniformly cont. on $[a, b]$.
Then, for $\epsilon > 0$, $\exists \delta > 0$ for any $x, y \in [a, b]$ s.t.

$$|x - y| < \delta \quad \text{implies} \quad |f(x) - f(y)| < \frac{\epsilon}{b - a}.$$

Let $P = \{a = t_0 < \cdots < t_n = b\}$ with $\mathrm{mesh}(P) < \delta$.
Then, $M(f, I_k) - m(f, I_k) < \epsilon/(b - a)$, so

$$U(f, P) - L(f, P) < \sum_{k=1}^{n} \frac{\epsilon}{b - a} \mu_k = \epsilon,$$

hence f is integrable on $[a, b]$. ■

· Operation between Integrals

: On this chapter, let f, g be integrable ftns on $[a, b]$.

- For $c \in \mathbb{R}$, cf is integrable, and $\int_a^b cf = c\int_a^b f$.

pf. ($\int_a^b cf = c\int_a^b f$)

Let $P = \{a = t_0 < \cdots < t_n = b\}$.

(i: $c > 0$) Let $c > 0$. Since

$$\begin{aligned} M(cf, I_k) &= \sup\{cf(x) : x \in I_k\} \\ &= c \cdot \sup\{f(x) : x \in I_k\} = c \cdot M(f, I_k), \end{aligned}$$

$$\begin{aligned} U(cf, P) &= \sum_{k=1}^{n} M(cf, I_k) = c\sum_{k=1}^{n} M(f, I_k) \\ &= cU(f, P), \end{aligned}$$

thus $U(cf) = cU(f)$, and similarly, $L(cf) = cL(f)$.
Since f is integrable, we conclude

$$U(cf) = cU(f) = c\int_a^b f = cL(f) = L(cf),$$

so cf is integrable, and $\int_a^b cf = c\int_a^b f$. □

(ii: $c = -1$) Since

$$\begin{aligned} M(-f, I_k) &= \sup\{-f(x) : x \in I_k\} \\ &= -\inf\{f(x) : x \in I_k\} = -m(f, I_k), \end{aligned}$$

$U(-f, P) = -L(f, P)$, thus

$$\begin{aligned} U(-f) &= \inf\{U(-f, P) : P \text{ is ptt. of } [a, b]\} \\ &= -\sup\{L(f, P) : P \text{ is ptt. of } [a, b]\} = -L(f). \end{aligned}$$

Similarly, $L(f) = -U(f)$.
Since f is integrable,

$$-U(f) = -L(f) = L(-f) = U(-f) = -\int_a^b f,$$

hence $-f$ is integrable, and $\int_a^b (-f) = -\int_a^b f$. □

(iii: $c < 0$) Let $c < 0$, then $c = -d$ for some $d > 0$,
then df and $(-d)f = -(df) = cf$ are int.ble on $[a, b]$
by (i), (ii), and

$$\int_a^b cf = \int_a^b (-df) = -\int_a^b df = (-d)\int_a^b f = c\int_a^b f.$$

■

- $f+g$ is integrable, and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

pf.

Let $\epsilon > 0$.

Integrability. Since f and g are integrable, there exist partitions P_1, P_2 of $[a, b]$ s.t.

$$\begin{cases} U(f, P_1) - L(f, P_2) < \epsilon/2, \\ U(g, P_2) - L(g, P_2) < \epsilon/2. \end{cases}$$

Let $P = P_1 \cup P_2$, then

$$\begin{cases} U(f, P) - L(f, P) < \epsilon/2, \\ U(g, P) - L(g, P) < \epsilon/2. \end{cases}$$

Since $m(f+g, S) \geqslant m(f, S) + m(g, S)$ f.a. $S \in [a, b]$ by the property of infimum, we have

$$L(f+g, P) \geqslant L(f, P) + L(g, P).$$

Similarly, we can derive

$$U(f+g, P) \leqslant U(f, P) + U(g, P).$$

Hence we conclude

$$\begin{aligned} &U(f+g, P) - L(f+g, P) \\ &\leqslant U(f, P) + U(g, P) - L(f, P) - L(g, P) \\ &< \epsilon/2 \cdot 2 = \epsilon, \end{aligned}$$

so $f+g$ is integrable on $[a, b]$. □

Equality. We calculate

$$\begin{aligned} \int_a^b (f+g) = U(f+g) &\leqslant U(f, P) + U(g, P) \\ &< L(f, P) + L(g, P) + \epsilon \\ &\leqslant L(f) + L(g) + \epsilon = \int_a^b f + \int_a^b g + \epsilon, \end{aligned}$$

$$\begin{aligned} \int_a^b (f+g) = L(f+g) &\geqslant L(f, P) + L(g, P) \\ &> U(f, P) + U(g, P) - \epsilon \\ &\leqslant U(f) + U(g) - \epsilon = \int_a^b f + \int_a^b g - \epsilon. \end{aligned}$$

Since ϵ is arbitrary, we have

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g.$$

■

· Inequalities on Integrals

- Preservation of Inequality

: For integrable ftn. f, g on $[a, b]$,

if $f(x) \leqslant g(x)$ for all $x \in [a, b]$, then $\int_a^b f \leqslant \int_a^b g$.

pf.

For integrable ftn $h = g - f (= g + (-1)f)$, $h(x) \geqslant 0$ for all $x \in [a, b]$.

Then $L(h, P) \geqslant 0$ for all partitions P of $[a, b]$,

thus $\int_a^b h = L(h) \geqslant 0$, so we conclude

$$\int_a^b g = \int_a^b f + \int_a^b h \leqslant \int_a^b f.$$

■

- For cont. nonnegative ftn g on $[a, b]$,

if $\int_a^b g = 0$, then $g(x) = 0$ for all $x \in [a, b]$.

pf.

Assume $g(x_0) \neq 0$ for some point x_0.

Then, $\exists \alpha > 0$ s.t. $g(x_0) \geqslant \alpha$, so $\exists [c, d] \subseteq [a, b]$ s.t. $g(x) \geqslant \alpha/2$ for all $x \in [c, d]$.

We conclude

$$\int_a^b g \geqslant \int_c^d g \geqslant \frac{\alpha}{2}(d-c) > 0, \quad \text{thus} \quad \int_a^b g \neq 0.$$

This contradicts to $\int_a^b g = 0$,

thus $g(x) = 0$ for all $x \in [a, b]$. ■

- Integrability of absolute function

: If f is int.ble on $[a, b]$, then so is $|f|$, and

$$\left| \int_a^b f \right| \leqslant \int_a^b |f|.$$

· Splitting/Combining Integrals

: For $a < c < b$, let f be a ftn. on $[a, b]$ s.t. int.ble on each $[a, c]$ and $[c, b]$.

Then f is int.ble on $[a, b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$.

(pf. on next page.)

· Integrating Finitely Inequal Function

: For an int.ble ftn. f on $[a, b]$, let g be a ftn. on $[a, b]$ s.t. $g(x) = f(x)$ except for finitely many $x \in [a, b]$.

Then g is int.ble, and $\int_a^b f = \int_a^b g$.

pf. ($|\int_a^b f| \leqslant \int_a^b |f|$)

Integrability. Let $S \subseteq [a, b]$, and $x_0, y_0 \in S$.
Then we have

$$\begin{aligned} |f(x_0)| - |f(y_0)| &\leqslant |f(x_0) - f(y_0)| \\ &\leqslant M(f, S) - m(f, S), \quad \cdots\cdots (1) \end{aligned}$$

thus

$$\begin{aligned} &\sup\{|f(x_0)| - |f(y_0)| : x_0, y_0 \in S\} \\ &= \sup\{|f(x_0)| : x_0 \in S\} - \inf\{|f(y_0)| : y_0 \in S\} \\ &= M(|f|, S) - m(|f|, S) \\ &\leqslant M(f, S) - m(f, S) \quad \text{(by (1))} \qquad \cdots\cdots (2) \end{aligned}$$

For $\epsilon > 0$, since f is int.ble,
$\exists$ a partition P of $[a, b]$ s.t. $U(f, P) - L(f, P) < \epsilon$.
By (2) and letting $S = [t_{k-1}, t_k]$ for $k = 1, 2, \ldots, n$,

$$U(|f|, P) - L(|f|, P) \leqslant U(f, P) - L(f, P) < \epsilon,$$

hence $|f|$ is integrable on $[a, b]$. □

Inequality. By Thm 33.4(i), and since

$$-|f(x)| \leqslant f(x) \leqslant |f(x)| \quad \text{for all } x \in [a, b],$$

$$-\int_a^b |f| \leqslant \int_a^b f \leqslant \int_a^b |f| \implies \left|\int_a^b f\right| \leqslant \int_a^b |f|. \qquad \blacksquare$$

pf. ($\int_a^b f = \int_a^c f + \int_c^b f$)

Integrability. Since f is bdd. on $[a, c]$ and $[c, b]$,
so is on $[a, b]$.
For $\epsilon > 0$, $\exists$ partitions P_1, P_2 of each $[a, c]$, $[c, b]$ s.t.

$$\begin{cases} U_a^c(f, P_1) - L_a^c(f, P_1) < \epsilon/2, \\ U_c^b(f, P_2) - L_c^b(f, P_2) < \epsilon/2 \end{cases}$$

since f is int.ble on both $[a, c]$ and $[c, b]$.

For $P = P_1 \cup P_2$, it is obvious that

$$\begin{cases} U_a^b(f, P) = U_a^c(f, P_1) + U_c^b(f, P_1), \\ L_a^b(f, P) = L_a^c(f, P_1) + L_c^b(f, P_2). \end{cases}$$

Then $U_a^b(f, P) - L_a^b(f, P) < \epsilon$, thus f is int.ble on $[a, b]$. □

Equality. We calculate

$$\begin{aligned} \int_a^b f &\leqslant U_a^b(f, P) = U_a^c(f, P_1) + U_c^b(f, P_2) \\ &< L_a^c(f, P_1) + L_c^b(f, P_2) + \epsilon \leqslant \int_a^c f + \int_c^b f + \epsilon, \end{aligned}$$

and similarly, $\int_a^b f \geqslant \int_a^c f + \int_c^b f - \epsilon$.
Since ϵ is arbitrary, we have $\int_a^b f = \int_a^c f + \int_c^b f$. $\blacksquare$

· Function Defined Piecewise

- Piecewise Monotonic Fuction

: A ftn. f on $[a, b]$ is piecewise monotonic
if $\exists P = \{a = t_0 < \cdots < t_n = b\}$ of $[a, b]$ s.t.
f is monotonic on each (t_{k-1}, t_k).

- Piecewise Continuous Function

: A ftn. f on $[a, b]$ is piecewise continuous
if $\exists P = \{a = t_0 < \cdots < t_n = b\}$ of $[a, b]$ s.t.
f is uniformly continuous on each (t_{k-1}, t_k).

- Expansion of Piecewise-defined Function

· For a bounded piecewise monotonic funtion f on $[a, b]$,
f can be expanded to a mono. ftn. $\tilde{f}$ on $[t_{k-1}, t_k]$.

· For a piecewise continuous funtion f on $[a, b]$,
f can be expanded to a cont. ftn. $\tilde{f}$ on $[t_{k-1}, t_k]$.

$$\left(\text{You can take } \tilde{f}(x) = \begin{cases} f(x) & \text{if } x \in (t_{k-1}, t_k), \\ \lim\limits_{x \to t_{k-1}} f(x) & \text{if } x = t_{k-1}, \\ \lim\limits_{x \to t_k} f(x) & \text{if } x = t_k. \end{cases}\right)$$

- Integrability

: If f is a bounded piecewise monotonic function on $[a, b]$,
or a piecewise continuous function on $[a, b]$,
then f is integrable on $[a, b]$.

pf. (monotonic)

Let f be a bounded piecewise monotonic ftn. on $[a, b]$
with a partition $P = \{a = t_0 < \cdots < t_n = b\}$.
Then f can be extended to a mono. ftn. f_k on $[t_{k-1}, t_k]$

$$\text{s.t. } f_k(x) = \begin{cases} f(x) & \text{if } x \in (t_{k-1}, t_k), \\ \lim\limits_{x \to t_{k-1}} f(x) & \text{if } x = t_{k-1}, \\ \lim\limits_{x \to t_k} f(x) & \text{if } x = t_k. \end{cases}$$

Then, by Thm 33.1, f_k is int.ble on $[t_{k-1}, t_k]$.
Since $f(x) = f_k(x)$ on $[t_{k-1}, t_k]$ except t_{k-1} and t_k,
f is int.ble on $[t_{k-1}, t_k]$ and $\int_{t_{k-1}}^{t_k} f = \int_{t_{k-1}}^{t_k} f_k$.
Then, f is int.ble on $[a, b]$ by Thm 33.6, and

$$\int_a^b f = \sum_{k=1}^n \int_{t_{k-1}}^{t_k} f. \qquad \blacksquare$$

§34. Fundamental Theorem of Calculus

· Fundamental Theorem of Calculus I

: For a function g s.t. cont. on $[a, b]$ & diff.ble on (a, b), if g' is int.ble on $[a, b]$, then

$$\int_a^b g' = \Big[g(x)\Big]_a^b = g(b) - g(a).$$

pf.

For $\epsilon > 0$, $\exists$ a partition $P = \{a = t_0 < \cdots < t_n = b\}$ s.t. $U(g', P) - L(g', P) < \epsilon$.
Apply the Mean Value Theorem to g on $[t_{k-1}, t_k]$:

$$\exists x_k \in (t_{k-1}, t_k) \text{ s.t. } g'(x_k) = \frac{g(t_k) - g(t_{k-1})}{t_k - t_{k-1}},$$

i.e., $g'(x_k)(t_k - t_{k-1}) = g(t_k) - g(t_{k-1})$.

Then $\sum_{k=1}^n g'(x_k)\Delta_k = \sum_{k=1}^n [g(t_k) - g(t_{k-1})] = g(b) - g(a)$.
Since $m(g', I_k) \leqslant g'(x_k) \leqslant M(g', I_k)$,

$$L(g', P) \leqslant \sum_{k=1}^n g'(x_k)\Delta_k = g(b) - g(a) \leqslant U'(g, P).$$

And we know $L(g', P) \leqslant \int_a^b g' \leqslant U(g', P)$, thus

$$\left|\int_a^b g' - \sum_{k=1}^n g'(x_k)\Delta_k\right| = \left|\int_a^b g' - (g(b) - g(a))\right| < \epsilon.$$

Since ϵ is arbitrary, we conclude $\int_a^b g' = g(b) - g(a)$. ■

· Integration by Parts

: For functions u, b s.t. cont. on $[a, b]$ & diff.ble on (a, b), if u' and v' are int.ble on $[a, b]$,

$$\int_a^b u'(t)v(t)\,\mathrm{d}t + \int_a^b u(t)v'(t)\,\mathrm{d}t = \Big[u(t)v(t)\Big]_a^b.$$

pf.

Let $g = uv$, where g is cont. on $[a, b]$ and diff.ble on (a, b), and $g' = u'v + uv'$ is int.ble. Then we conclude

$$\int_a^b u'v + \int_a^b uv' = \int_a^b g' = g(t)\Big|_a^b = u(t)v(t)\Big|_a^b.$$

■

· Fundamental Theorem of Calculus II

: For an int.ble ftn. f on $[a, b]$ and $x \in [a, b]$, let

$$F(x) = \int_a^x f(t)\,\mathrm{d}t.$$

Then F is cont. on $[a, b]$.

If f is cont. at $x_0 \in (a, b)$, then F is diff.ble at x_0 and

$$F'(x_0) = f(x_0).$$

pf.

Continuity. Since f is int.ble, it is bdd, i.e.,
$\exists B \geqslant 0$ s.t. $|f(x)| \leqslant B$ for all $x \in [a, b]$.
For $\epsilon > 0$, let $\delta = \epsilon/B$, and $x, y \in [a, b]$ s.t. $0 < y - x < \delta$.
Then we have

$$\begin{aligned}|F(y) - F(x)| &= \left|\int_a^y f - \int_a^x f\right| = \left|\int_x^y f(t)\,\mathrm{d}t\right| \\ &\leqslant \int_x^y |f| \leqslant \int_x^y B\,\mathrm{d}t = B(y - x) < \epsilon,\end{aligned}$$

hence F is [unif.] cont. on $[a, b]$ □

. . . . . .

Equality. Let $x \in (a, b) \setminus \{x_0\}$. Note that

$$\frac{F(x) - F(x_0)}{x - x_0} = \frac{\int_{x_0}^x f(t)\,\mathrm{d}t}{x - x_0}, \quad f(x_0) = \frac{\int_{x_0}^x f(x_0)\,\mathrm{d}t}{x - x_0}.$$

Then we have

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x) = \frac{1}{x - x_0}\int_{x_0}^x (f(t) - f(x_0))\,\mathrm{d}t.$$

Let $\epsilon > 0$. Since f is cont. on x_0, $\exists \delta > 0$ s.t.

$$x \in (a, b) \text{ and } |t - x_0| < \delta \quad \text{imply} \quad |f(t) - f(x_0)| < \epsilon.$$

Then, for $x \in (a, b)$ s.t. $0 < |x - x_0| < \delta$,

$$\begin{aligned}\left|\frac{F(x) - F(x_0)}{x - x_0} - f(x)\right| &= \left|\frac{1}{x - x_0}\int_{x_0}^x (f(t) - f(x_0))\,\mathrm{d}t\right| \\ &\leqslant \frac{1}{|x - x_0|}\int_{x_0}^x |f(t) - f(x_0)|\,\mathrm{d}t \\ &\leqslant \frac{1}{|x - x_0|}\int_{x_0}^x \epsilon\,\mathrm{d}t < \epsilon,\end{aligned}$$

hence $\lim_{x \to x_0} \frac{F(x) - F(x_0)}{x - x_0} = F'(x_0) = f(x)$. ■

· Change of Variable $(a, b \in J)$

: For a diff.ble ftn. u on an open interval J s.t. u' is cont., let I be an open interval s.t. $u'(J) \subseteq I$. If f is cont. on I,

$$\int_a^b (f \circ u)(x)u'(x)\,\mathrm{d}x = \int_{u(a)}^{y(b)} f(u)\,\mathrm{d}u.$$

pf.

Fix $c \in I$, and let $F(a) = \int_c^a f$. Then, $F' = f$ on I by FTCII.
Let $g = F \circ u$, then g is diff.ble on J, and $g' = (f \circ u)u'$. Then,

$$\begin{aligned}\int_a^b (f \circ u)u' &= \int_a^b g' = g(x)\Big|_a^b = F(u(x))\Big|_a^b \\ &= \int_c^{u(b)} f - \int_c^{u(a)} f = \int_{u(a)}^{u(b)} f.\end{aligned}$$

■

· Integral of Inversive Function

: For injective diff.ble ftn. g on I,

$$\int_a^b g + \int_{g(a)}^{g(b)} g^{-1} = xg(x)\Big|_a^b.$$

pf.

$$\begin{aligned}\int_a^b (g^{-1} \circ g)(x)g'(x)\,\mathrm{d}x &= \int_{g(a)}^{g(b)} g^{-1} \\ = \int_a^b xg'(x)\,\mathrm{d}x \qquad &= xg(x)\Big|_a^b - \int_a^b g.\end{aligned}$$

■